Unit 3 – While Loops in p5.js

Name: _____ Date: _____ Period: _____

Syntax Rules: Read Before You Start

Every while loop in p5.js has the same shape and the same four parts:

__start_;

while (conditon){repeat; update;}

a **start** before the loop, the keyword `while`, the **question:condition** inside parentheses ( ), then the **body:repeat** and the **update:step** inside curly braces { }.

```
let x = 0;                  // start: before the loop
while (x < 800) {           // the question: keep going WHILE true
  circle(x, 300, 20);       // the body: the work
  x = x + 40;               // the update: moves x toward false
}
```

AP exam version: `REPEAT UNTIL (x ≥ 800) { … }` keeps going UNTIL true, so its condition is the **opposite** of the while.

Every question uses an 800 × 600 canvas, so `width` = 800 and `height` = 600.

Section 1: Syntax Rules Table

Each line below has one mistake. Read why it is wrong, then write the corrected code in the Correct column.

Wrong	Why it's wrong	Correct
`while x < 800 { }`		
`while (x < 800); { }`		

`while (x < 800) {`
`  circle(x, 100, 20);`
`}`		

`while (x < 800) {`
`  let x = 0;`
`  x = x + 40;`
`}`		

`while (x < 800) {`
`  x + 40;`
`}`		

`while (x < 800) {`
`  x = x - 40;`
`}`		

`while (x = 800) { }`		
`while (x < 800) { }` `x = x + 40;`		
`REPEAT UNTIL (x > 800)` `→ while (x > 800)`		

Section 2: What's the Value? (Trace It Yourself)

Trace each block of code. Write the final value of each variable. Make a trace table (x | question | true/false) in the margin if it helps.

1. Count the trips. What are count and x after the loop?

```
let x = 0;
let count = 0;
while (x < 100) {
  x = x + 25;
  count = count + 1;
}
```

count = _____ x = _____

4. Zero times?

```
let x = 900;
let bulbs = 0;
while (x < 800) {
  bulbs = bulbs + 1;
  x = x + 40;
}
```

bulbs = _____ x = _____
Why? _____

5. Doubling.

```
let n = 1;
let trips = 0;
while (n < 50) {
  n = n * 2;
  trips = trips + 1;
}
```

n = _____ trips = _____

6. Ends or never ends? Write ENDS (and how many trips) or INFINITE.

a. `let x = 0; while (x < 800) { x = x - 20; }`

_____

b. `let x = 0; while (x != 100) { x = x + 30; }`

_____

c. `let x = 0; while (x != 90) { x = x + 30; }`

_____

7. Crossy road lanes.

```
let y = 100;
let lanes = 0;
while (y < 600) {
  rect(0, y, 800, 60);
  lanes = lanes + 1;
  y = y + 120;
}
```

lanes = _____ y = _____

8. AP exam pseudocode.

```
x ← 2
REPEAT UNTIL (x > 20)
{
  x ← x * 3
}
```

x = _____ How many times did the body run? _____

Section 3: Write It Yourself

Write each loop by hand. Check every answer against the Syntax Rules table: a start before the loop, parentheses around the question, braces around the body, and an update that moves toward false.

Variables available: `x`, `y`, `bulbs`, `flakes`, `lanes`, `lives`, `ax`, `ay`, `tries`, the ship at `(400, 300)`, and `width` (800) and `height` (600).

1. Holiday lights. Start x at 40. While x is less than width, draw circle(x, 80, 20), then move x 60 to the right.

---

2. Count the lights. Add to your answer for #1 so bulbs count every circle. After the loop, show text("bulbs " + bulbs, 20, 30). (Start bulbs at 0.

---

3. Snowfall. Draw 200 snowflakes, circle(random(width), random(500), 4), counting them with flakes.

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4. Crossy road lane line. Draw dashes rect(x, 300, 40, 8) starting at x = 20, one every 80 pixels, all the way across the canvas.
